ARTIFICIAL INTELLIGENCE MINI-PROJECT

Brain-Inspired AI for Real-Time Heart Rhythm Monitoring

Accurate, Low-Power Arrhythmia Classification using Spiking Neural Networks (SNNs)

Author: Ketan Shinde | Domain: Neuromorphic AI & Healthcare Machine Learning

Problem Statement & Clinical Motivation

The Medical Need

- Heart diseases are the leading cause of mortality worldwide.
- Irregular heart rhythms (arrhythmias) can happen randomly and without warning.
- Patients need continuous 24/7 monitoring through small wearable patches.
- Catching dangerous heartbeats early prevents sudden cardiac arrest.

The Problem with Traditional AI

- Standard deep learning models (like CNNs or LSTMs) do massive continuous calculations.
- They consume heavy battery power, draining wearable monitors in just a few hours.
- Sending raw heart signals to the cloud introduces lag and risks patient privacy.
- We need an AI that can think locally on the wearable device in real time.

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Understanding Spiking Neural Networks (SNNs)

1. Brain-Inspired Pulses

• In our brains, neurons communicate using tiny electrical pulses called spikes.

• Instead of sending heavy decimal numbers, SNN neurons send simple on/off signals (0 or 1).

• Information is carried in the rhythm and timing of when these pulses fire.

2. The Leaky Bucket Analogy

• Think of each artificial neuron like a bucket with a small leak at the bottom.

• Incoming pulses add water into the bucket (Integrate).

• If nothing arrives, the level slowly drops over time (Leak).

• When the bucket fills to the top, it fires a pulse and resets to empty (Fire & Reset).

3. Why It Saves Power

• Neurons stay completely silent until an actual pulse arrives.

• No spikes = no activity = zero power consumed.

• Instead of complex multiplications, the network just does simple additions when a pulse hits.

• Ideal for smart, battery-powered medical wearables.

The Dataset: MIT-BIH Arrhythmia Benchmark

Benchmark: MIT-BIH Arrhythmia Database (PhysioNet)

Standard clinical benchmark with over 100,000 real patient heartbeats mapped to the 4 official AAMI medical categories:

Normal (N)

90,113 beats (89.5%)

Healthy sinus rhythm and typical heart patterns.

Supraventricular (S)

2,781 beats (2.8%)

Abnormal beats starting early in the upper heart chambers (atria).

Ventricular (V)

7,008 beats (7.0%)

Dangerous irregular contractions in the lower chambers (PVCs). Critical!

Fusion (F)

802 beats (0.8%)

A rare, complex blend of normal and ventricular heartbeats.

Data Pipeline: From Raw Heartbeat to Brain Spikes

1. Slicing the Heartbeat

• We locate each heartbeat by finding its main sharp spike (the R-peak).

• We crop a neat window of 128 points around it.

• This captures the complete heartbeat wave from start to finish.

• Ensures every beat fed into the AI has identical length.

2. Cleaning the Signal

• Real heart signals get noisy when patients breathe, walk, or move.

• We apply a robust scaling filter that smoothly cleans up baseline drift.

• Scales every heartbeat neatly into a uniform 0 to 1 range.

• Ensures signals look clean and consistent for the neural network.

3. Converting to Spikes

• How does an SNN read a heartbeat? Using 'Rate Coding' across 25 time steps.

• Think of it like Morse code:

• • Tall peaks (QRS wave) → rapid, dense bursts of spikes.

• • Flat baseline → silent, almost zero spikes.

• Translates the heart signal into a rhythmic language the AI understands.

Network Architecture: Simple & Ultra-Compact

Layer Structure: 128 → 32 → 32 → 4

• Input Layer (128 channels): One input for each point in the heartbeat window.

• Hidden Layer 1 (32 neurons): Detects basic wave shapes and early rhythm patterns.

• Hidden Layer 2 (32 neurons): Combines patterns to spot specific arrhythmia traits.

• Output Layer (4 neurons): One neuron for each category (Normal, Supraventricular, Ventricular, Fusion).

• Winner Takes All: Whichever output neuron fires the most spikes is our final prediction!

Why the Design is So Efficient

✓ Tiny Memory Footprint: Total network size is only 5,248 parameters (roughly 5.2 Kilobytes!).

✓ Runs on Simple Microchips: Can easily fit into tiny, ultra-cheap microcontrollers or wearable chips without external memory.

✓ No Heavy Math: Calculations use simple additions rather than thousands of floating-point multiplications.

✓ Fast Response: Analyzes each heartbeat in a fraction of a millisecond.

Training the AI: Overcoming the Spike Problem

The Challenge: Spikes Don't Have Slopes

- Modern AI normally learns using calculus slopes (gradients).
- When an AI makes an error, it follows the slope backwards to tweak its weights.
- The Problem with Spikes: A spike is an instant jump from 0 to 1.
- Because it is an instant cliff, the slope is flat everywhere—which normally halts AI learning completely!
- For decades, this made training spiking networks with deep learning nearly impossible.

The Solution: Surrogate Gradients

✓ The Clever AI Trick: We separate forward prediction from backwards learning.

✓ When running forward: The neurons use real, discrete binary spikes.

✓ When learning backwards: The computer pretends the sharp cliff is a smooth, gentle hill.

✓ This allows standard optimizers (like Adam) to calculate corrections smoothly.

✓ Result: We get real biological spikes with the training power of modern deep learning!

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Teaching the Rhythm: Timing-Aware Learning

Why Standard Loss Fails

• Normal classification loss only checks: 'Did the model pick the right label at the end?'

• It completely ignores how quickly or steadily the pulses fired.

• Cardiac diagnosis needs timing awareness, not just a final vote.

Teaching Rhythm & Timing

• We used a timing-aware loss function (Van Rossum distance).

• The model converts spike pulses into smooth activity waves over time.

• It compares its predicted rhythm against the correct rhythm at every single time step.

• Penalizes both missing a beat and firing too late.

The Training Result

• Smooth and reliable learning across all 60 training epochs.

• Total error dropped smoothly by ~71% from start to finish.

• The network learned to fire quickly and decisively when an arrhythmia appears.

Experimental Results: Multi-Seed Benchmark

Model Version	Overall Accuracy	Balanced Score (F1)	Normal Detection	Arrhythmia Precision
Published SparrowSNN Baseline	93.26 ± 0.82%	71.14 ± 1.90%	93.4%	~50% (High false alarms)
Our Optimized SNN AI Model 🚀	**97.24 ± 0.19%**	**83.85 ± 1.36%**	**98.57%**	**~75% (Accurate & reliable)**

97.24% ± 0.19%

Overall Accuracy

+3.98% over standard baseline

83.85% ± 1.36%

Macro-F1 Score

+12.71% balanced improvement

± 0.19%

Super Low Variance

Consistently high across random seeds

5.2 KB

Total AI Size

Fits on any microcontroller

Clinical Impact: Slashing False Alarms

Test Confusion Matrix (Over 20,000 Beats)

	Pred N	Pred S	Pred V	Pred F
True N:	17,805	79	118	21
True S:	165	363	28	0
True V:	49	4	1,339	10
True F:	24	0	14	122

Key Highlights:
• Healthy Beat Accuracy: 98.79% (Rarely causes false alarms)
• Dangerous PVC Detection: 95.51% (Catches life-threatening beats)
• High Arrhythmia Precision: ~75% to 81% true alert rate

Why This Matters in Healthcare

• Cutting False Alarms by ~46%: Drops nuisance alarms on healthy beats by nearly half, directly solving 'alarm fatigue' for nurses and doctors.

• Clinical Trust: When this system sounds an alarm, doctors know there is a 75-80% probability it is genuine, compared to only ~50% in prior models.

• Critical Sensitivity: Catches premature ventricular contractions with over 95.5% reliability, preventing progression to fatal ventricular fibrillation.

• Edge-Ready: All of this happens instantly on the patient's device.